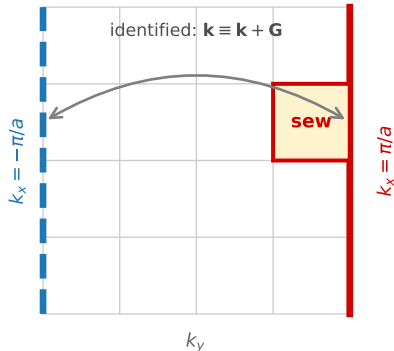


(a)

BZ plaquette at boundary

**(b)**

Sewing relation

Continuum / PWE:

$$u_{n,\mathbf{k}+\mathbf{G}}(\mathbf{r}) = e^{-i\mathbf{G}\cdot\mathbf{r}} u_{n,\mathbf{k}}(\mathbf{r})$$

or equivalently:

$$C_{\mathbf{G}',\mathbf{k}+\mathbf{G}} = C_{\mathbf{G}'+\mathbf{G},\mathbf{k}}$$

(PWE Fourier coefficients
shift by $\mathbf{G}$)**(c)**

When is sewing needed?

Continuum / PWE / FEM

$H(\mathbf{k})$ not periodic in $\mathbf{k}$
 $\Rightarrow$ sew at BZ boundary

Lattice model (e.g. QWZ)

$H(\mathbf{k})$ periodic by construction
 $\Rightarrow$ no sewing needed

*Energy-metric link variables
 must also be sewn for Maxwell*